

Experiment 12: Dining Philosophers Problem

C Program

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#define N 5
sem_t chopstick[N];

void *philosopher(void *num){
    int id=*(int*)num;
    printf("Philosopher %d is Thinking\n",id);
    sem_wait(&chopstick[id]);
    sem_wait(&chopstick[(id+1)%N]);
    printf("Philosopher %d is Eating\n",id);
    sleep(1);
    sem_post(&chopstick[id]);
    sem_post(&chopstick[(id+1)%N]);
    printf("Philosopher %d Finished Eating\n",id);
    pthread_exit(NULL);}

int main(){
    pthread_t thread[N]; int id[N];
    for(int i=0;i<N;i++) sem_init(&chopstick[i],0,1);
    for(int i=0;i<N;i++){ id[i]=i;
pthread_create(&thread[i],NULL,philosopher,&id[i]); }
    for(int i=0;i<N;i++) pthread_join(thread[i],NULL);
    for(int i=0;i<N;i++) sem_destroy(&chopstick[i]);
    return 0; }
```

Sample Output

```
Philosopher 0 is Thinking
Philosopher 1 is Thinking
Philosopher 2 is Thinking
Philosopher 3 is Thinking
Philosopher 4 is Thinking

Philosopher 0 is Eating
Philosopher 2 is Eating
Philosopher 0 Finished Eating
Philosopher 1 is Eating
Philosopher 2 Finished Eating
Philosopher 3 is Eating
Philosopher 1 Finished Eating
Philosopher 4 is Eating
Philosopher 3 Finished Eating
Philosopher 4 Finished Eating
```

Result

The Dining Philosophers problem was successfully simulated using threads and semaphores.